Elder Health Assistant

Inclusive Digital Technologies
Academic year 2025/2026

Designed4U

Our Team – Designed4U



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Digital and Interaction Design

Exclusion Barriers



Cognitive **decline** and **autonomy loss**



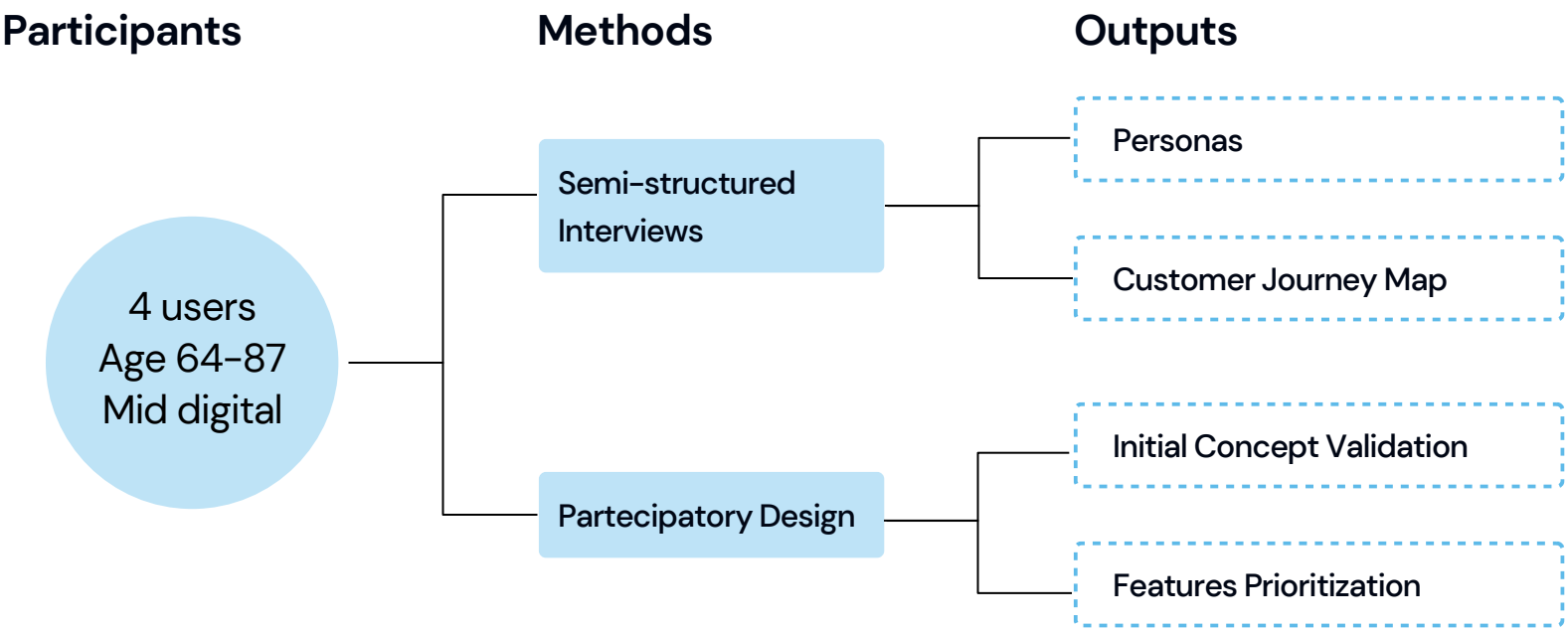
Poor **leaflets legibility**



Information Complexity & **Overload**

"When patients can't understand or remember, they can't be autonomous."

User Research



Personas

Low Vision

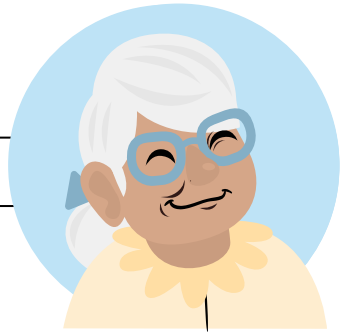
Struggle with small text

Habitual

Fixed daily habits

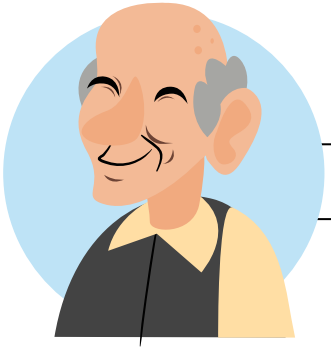
Self Reliant

Values indipendence



Concetta

72 years old, retired seamstress



Ettore

75 years old, retired teacher

Forgetful

Loses track of details

Needy

Needs support

Sociable

Enjoy familiar company

Customer Journey Map



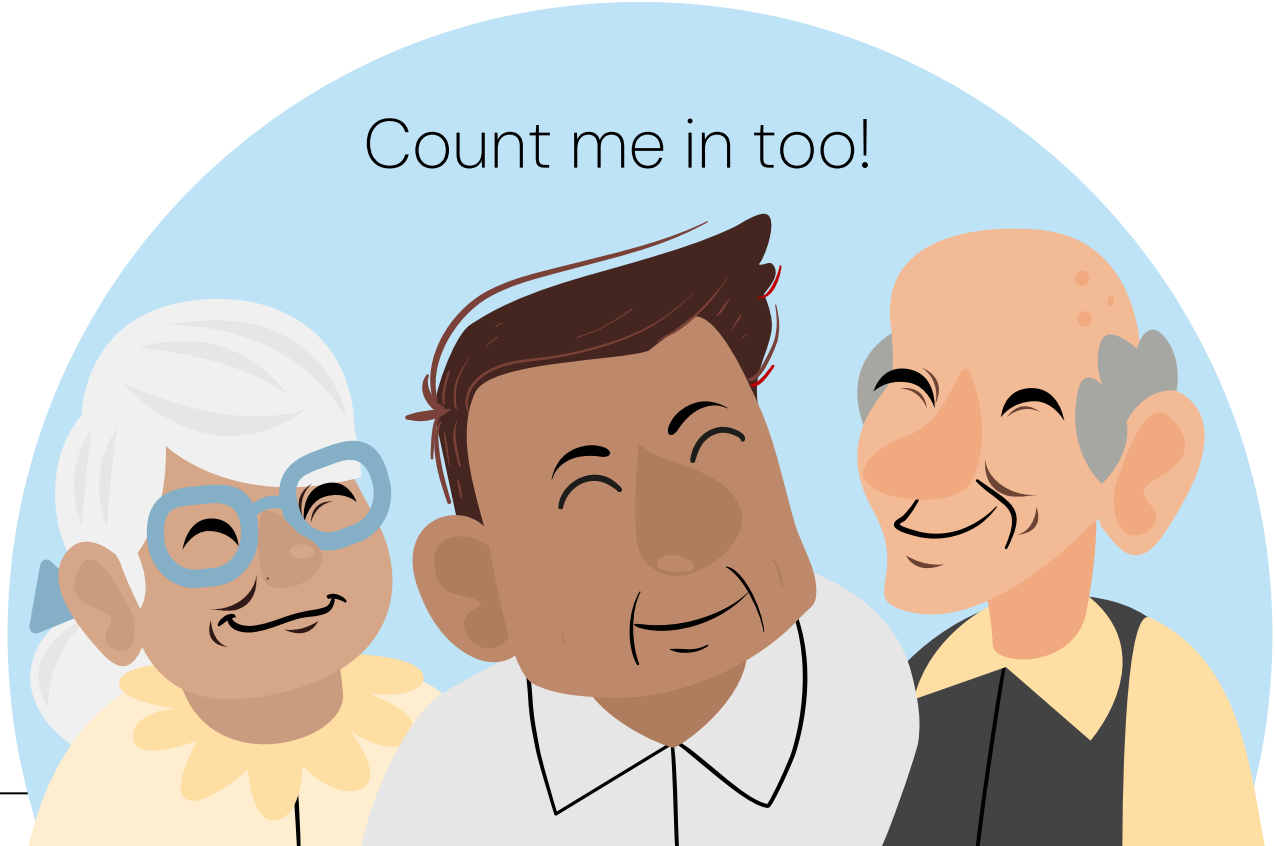
Prescription	Understanding	Preparation	Administration
Meeting with the **doctor**, and prescription of a **new medication**	**Reading the leaflet**, online research, **asking family** for help	**Organisation**: paper **lists**, memory, alarms, mixed methods	Correct medication intake at the **scheduled time**

User pain points

Frequent doubts & fears	Cognitive overload	Lack of support tools	Fragmented organization	Forgetfulness and errors
Need for **security** and **assistance**	Need for **readability** and **simplicity** of informations	Need to **feel autonomous** but supported	Need for **structured organization** strategies	Need for constant support and efficient **reminders**

User needs

The caregiver role



Caregiver needs

“The caregiver needs to stay informed and reassured, without disrupting the older person’s indipendence.”



Participatory design

Features presented

Voice Interaction

Text Reminders

Health App Sync

Box medication Scan

Doctor Visit Transcription

Medication Q&A

Visual Reminders

Easy to read leaflet

Your medicines – List

Photo Pill Verification

Participatory design

Features selected

Voice Interaction

Text Reminders

Health App Sync

Box medication Scan

Doctor Visit Transcription

Medication Q&A

Visual Reminders

Easy to read leaflet

Your medicines – List

Photo Pill Verification

Our Idea

Telegram Chatbot

Medicine Reminder

AI Assistant

Prototype

Caregiver connection

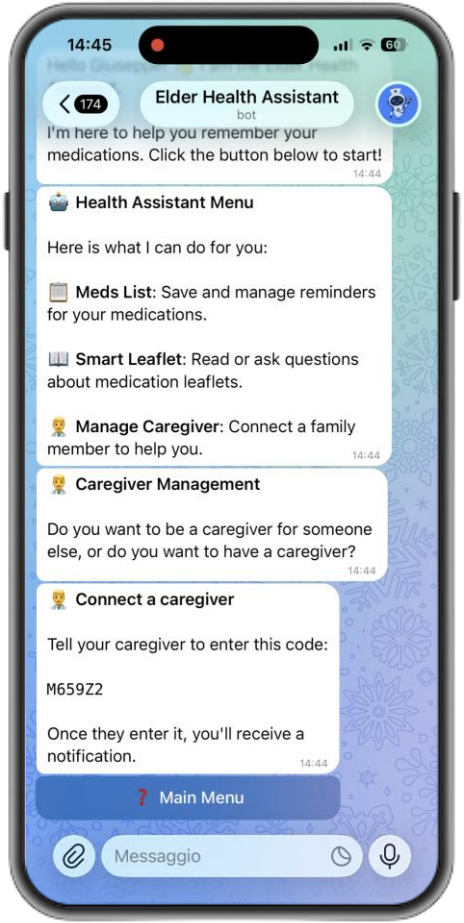
The user chooses to set his caregiver and open the connection generating the code

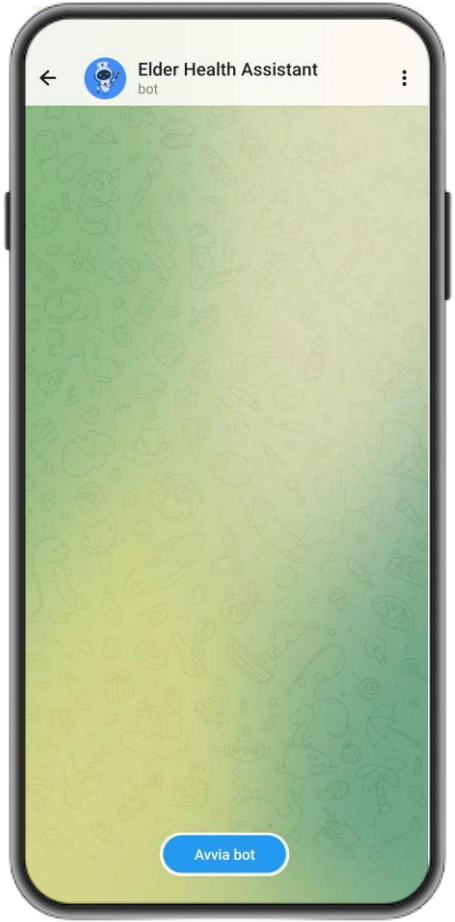


Prototype

Caregiver connection

The caregiver access to his user’s connection entering the code

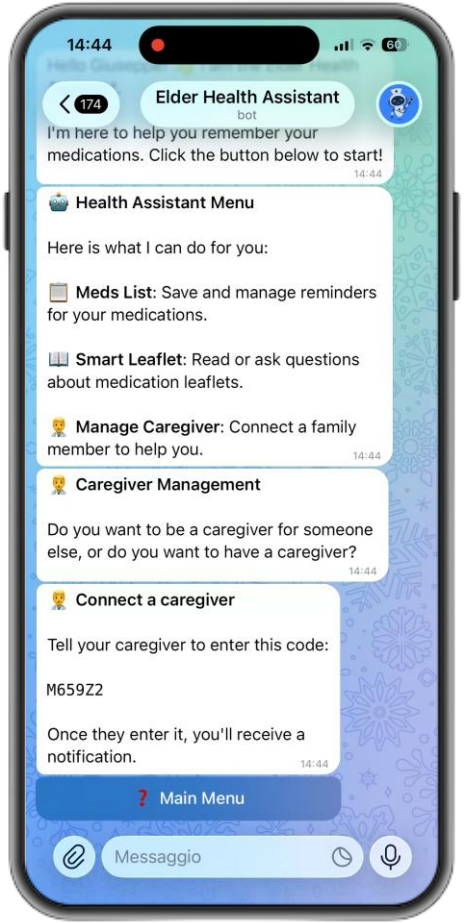


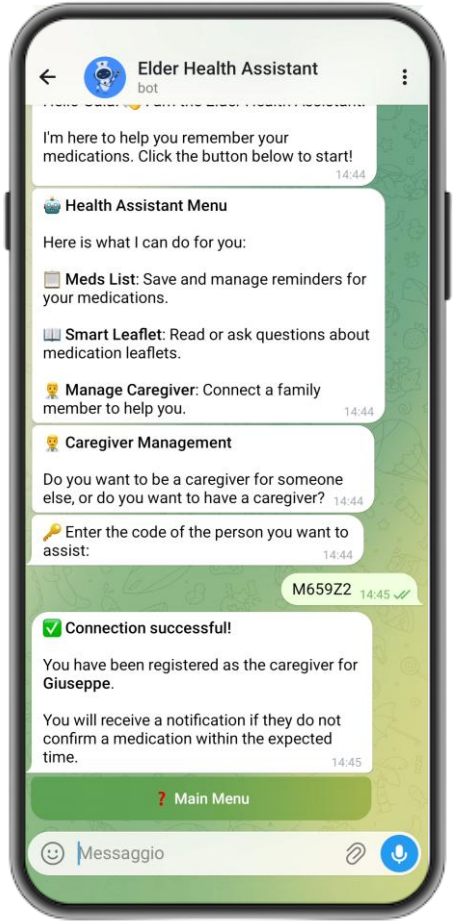


Prototype

Caregiver connection

The user receive the confirm when the caregiver is connected



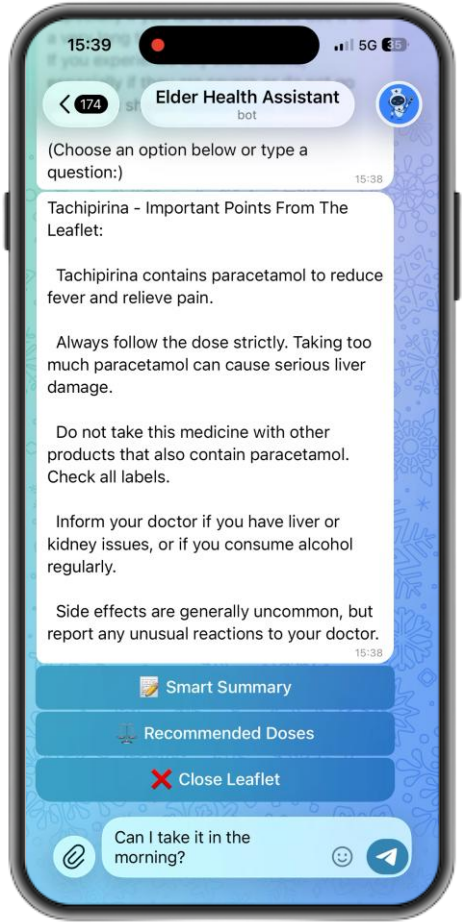


Prototype

Leaflet and AI chat

Open the leaflet of a specified medication in a easy-to-read version

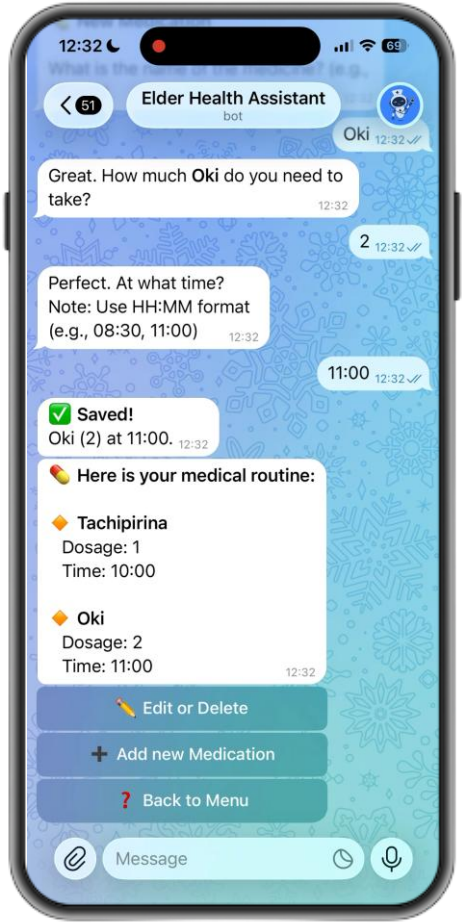
Choose between suggested interaction or ask any kind of question regarding the opened leaflet



Prototype

Reminders

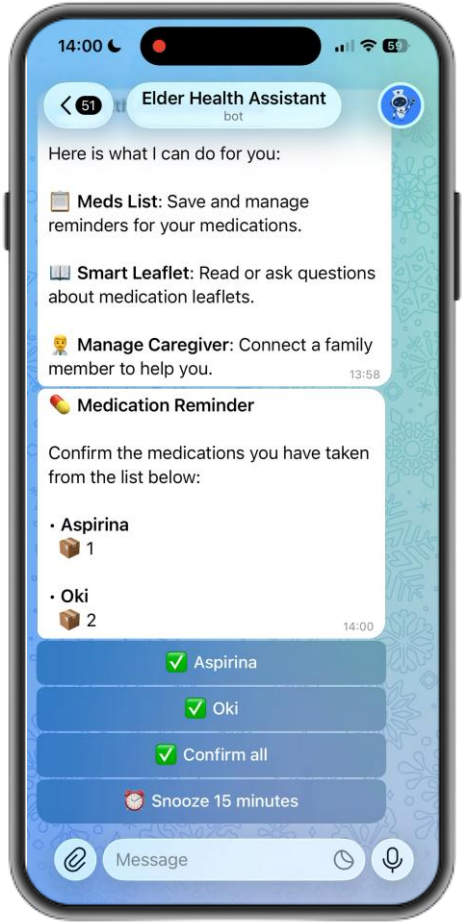
See your Meds List and add new medications to it to set new reminders



Prototype

Reminders

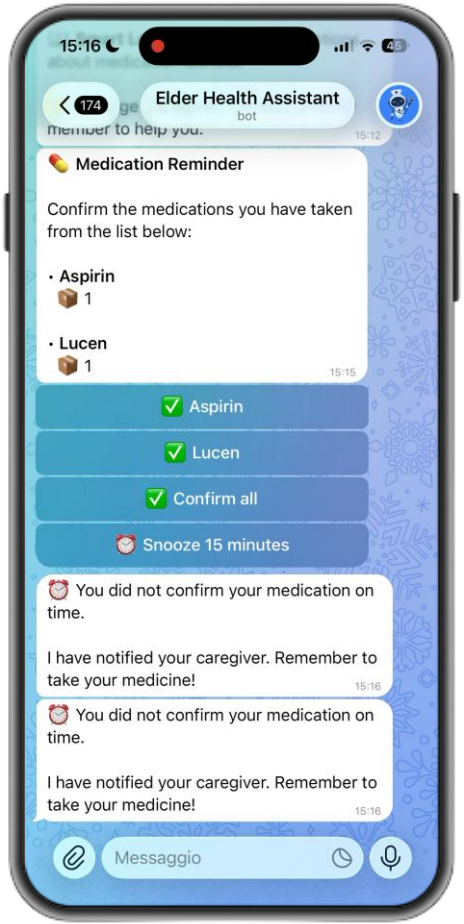
Multiple medications' reminder can be confirmed with a single button

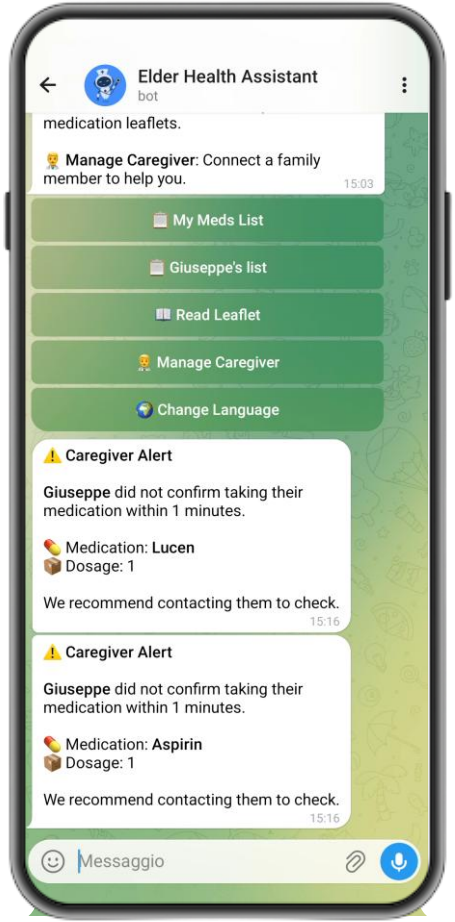


Prototype

Reminders

In the case the user forget to take medications and confirm it, the Caregiver is notified





Nielsen Heuristics

01 Visibility of system status

02 Match between system and real world

03 User control and freedom

04 Consistency and standards

05 Error Prevention

06 Recognition Rather than Recall

07 Flexibility and Efficiency of Use

08 Aestetic and Minimal Design

09 Help users with Errors

10 Help and documentation

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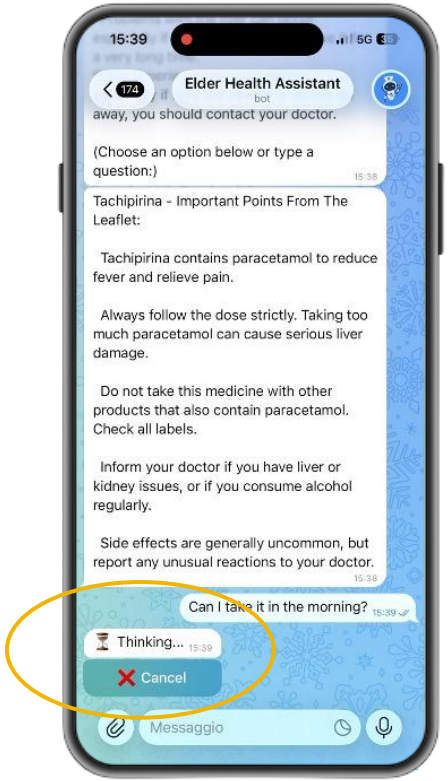
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AI is thinking

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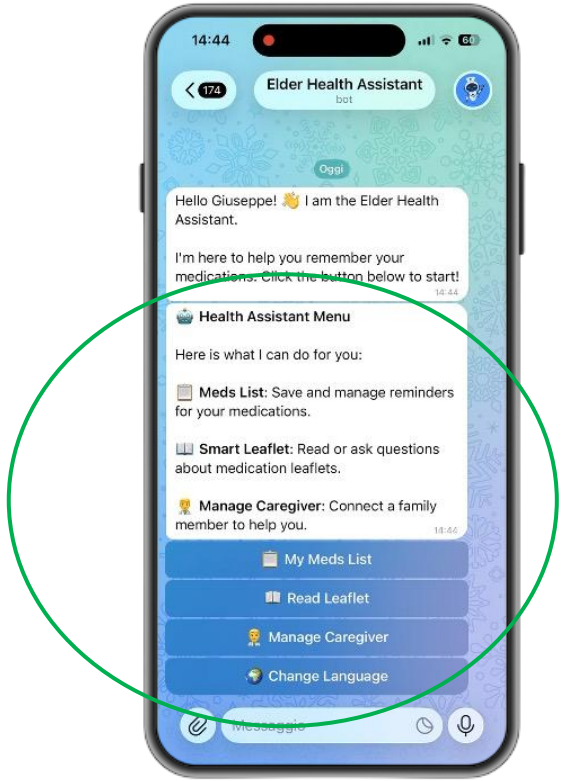
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Emoji and Menu

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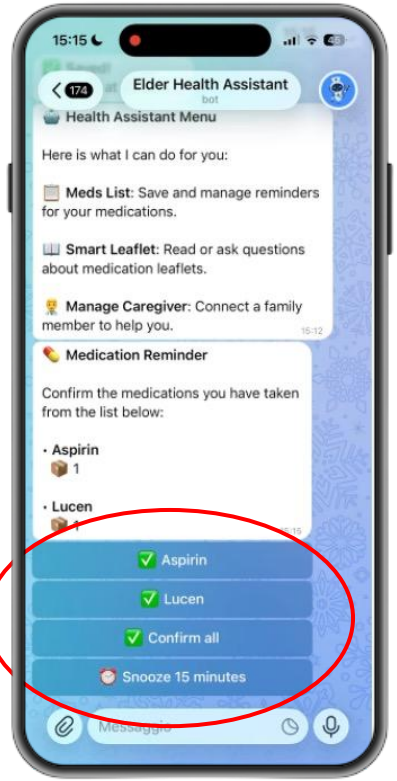
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Snooze and Confirmation

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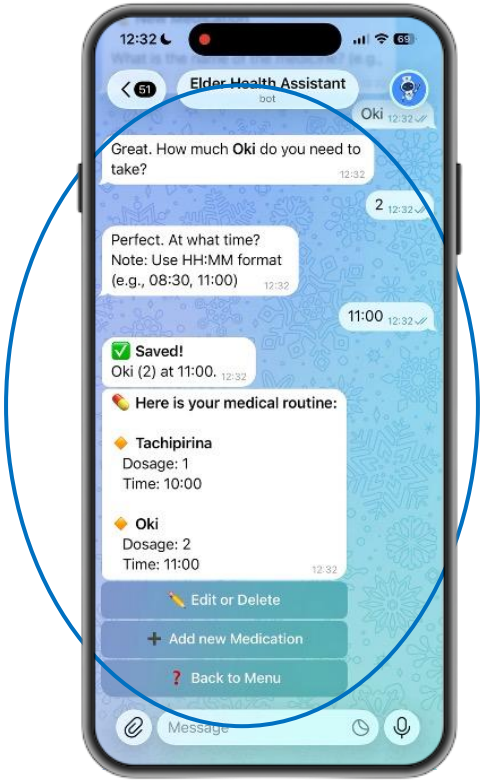
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Long Chat History

Nielsen Heuristics

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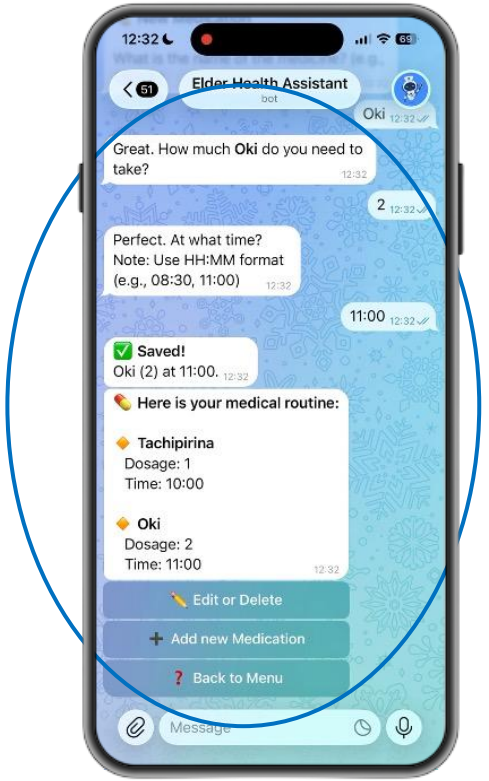
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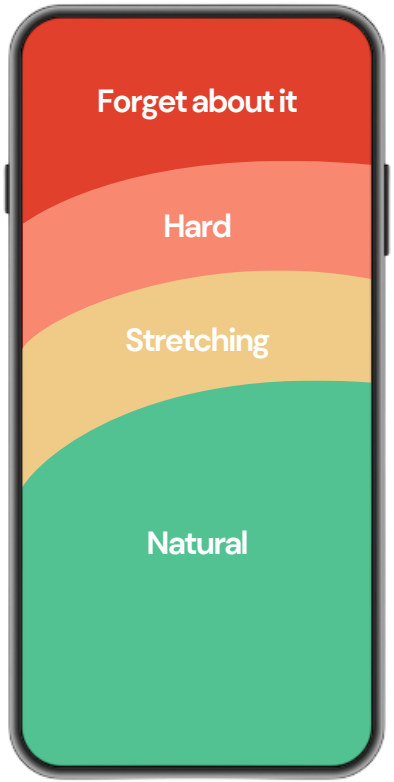
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Long Chat History



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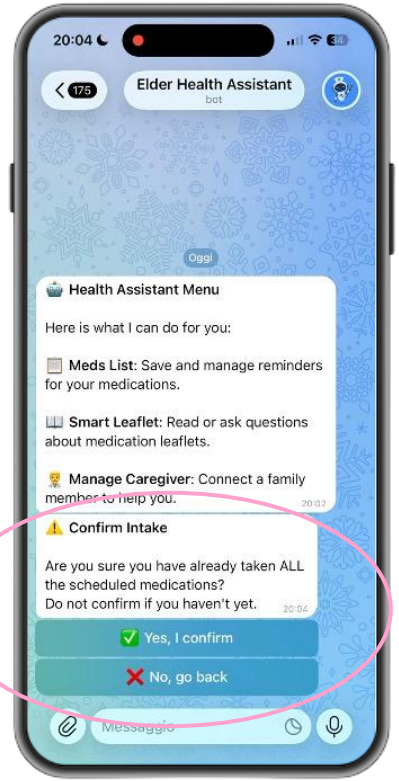
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Double Confirmation

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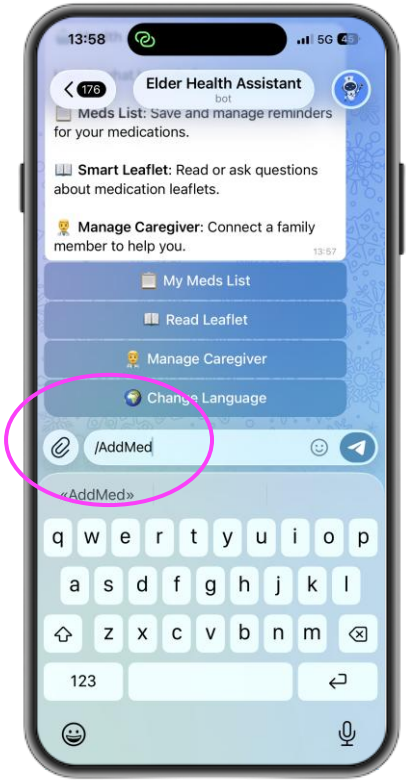
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Shortcut

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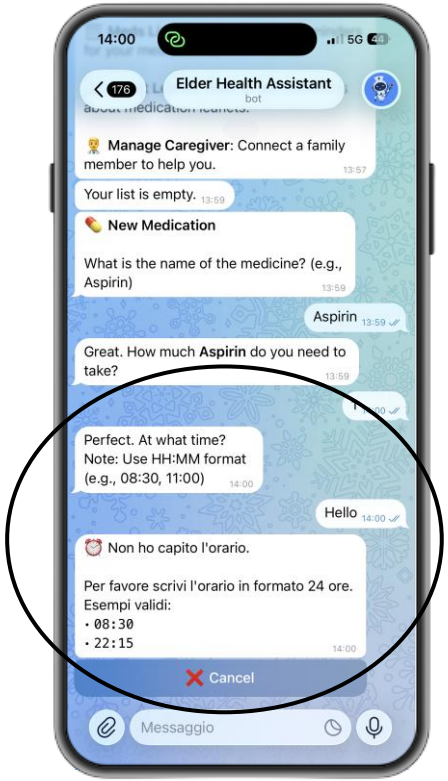
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Incorrect Format

Feedbacks

From Reviews

From problems...		**...To solutions**
Lack of **error recovery** for edge cases	→	**Double confirmation** mechanism to prevent errors
Alert fatigue for multi-medication users	→	**"Confirm All"** batch action to optimize daily routine

Feedbacks

From Heuristic Evaluation

From problems...		**...To solutions**
Visual clutter for the buttons	→	Consolidating all action buttons to a single **bottom anchor**
Users **cannot delay** notifications if busy	→	Introducing a "**Snooze**" Option

User Testing Validation

5 User Participants

3 Task Scenarios

TAM & Feedback



User Testing Validation

3 Task Scenarios



Reminder

‘’ You have started taking 1 pill of Tachipirina. Set a daily reminder for 8:00 AM ”

Leaflet

‘’ You are taking Tachipirina and want to consult the leaflet to check its main side effects ”

Caregiver

“ Your child wants to connect to your account to help you manage your medications. Allow access ”

Qualitative Feedback

" It works well. I would put a voice option too, reading is not so easy at my age "

" It makes me feel safe that my son can see when I forget my pills "

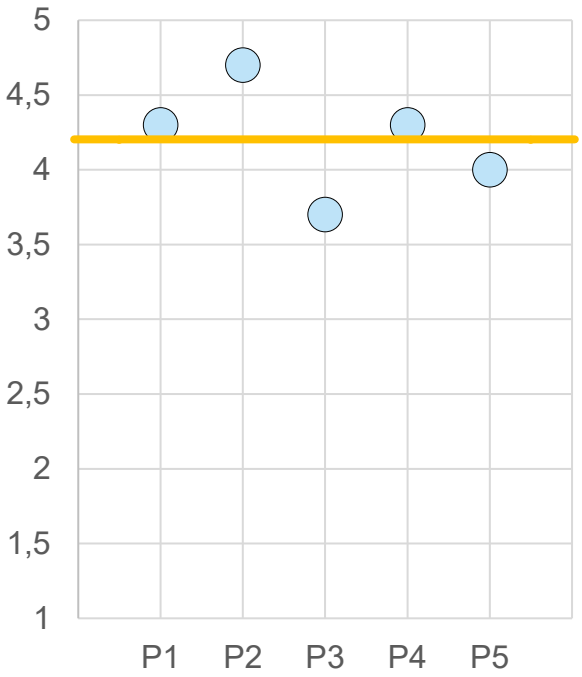
" Asking about medicines is very handy, I never liked reading those leaflets "

" I never used Telegram before, so at the start I was a bit lost "

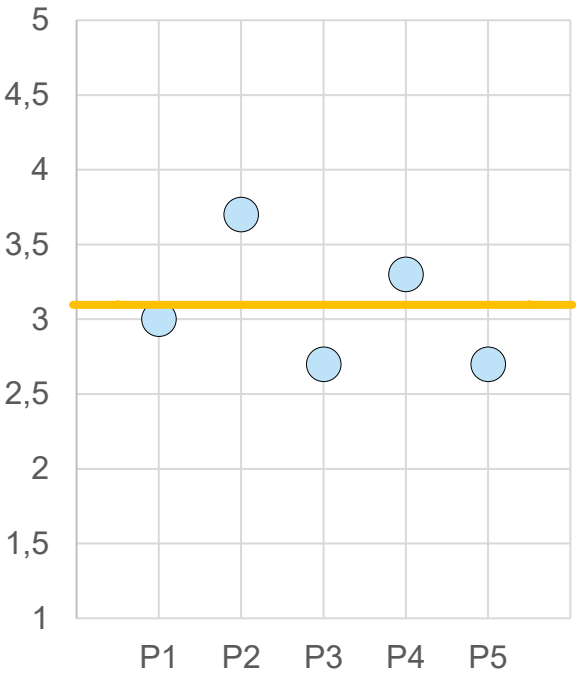


TAM Results

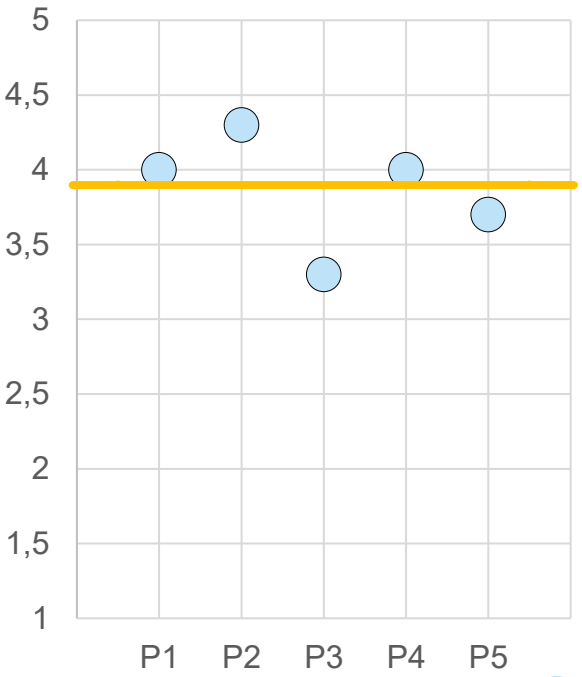
Perceived Usefulness



Ease of Use



Behavioral Intention



Future work

Introduce voice mode

01

Alexa integration

02

Add meds via photos

03

Invite more caregivers

04

Thank you!

Do you have any questions?

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Filippo Morelli
Vincenzo Picone
Gaia Scherini

